

Power-Law Covariance

A simplifying structure for data and representations

Princeton 2026 Summer School in Machine Learning

Misha Gromov

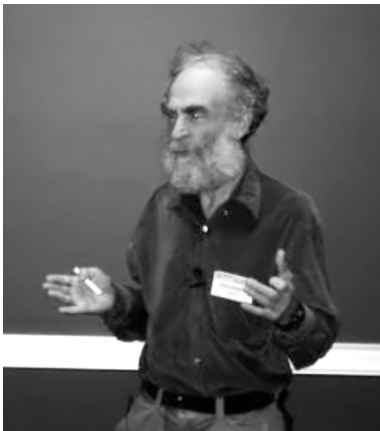


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Here the structure is the **covariance eigenspectrum**.

Covariance and principal directions

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Covariance is a **linear** statistic, but in high dimension it can still carry rich geometry. Later we will ask whether nonlinear network transformations weakly preserve that geometry.

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Faster decay; fewer visible directions.

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Example: Matérn kernels connect α to smoothness and intrinsic dimension.

Image spectra survive random layers

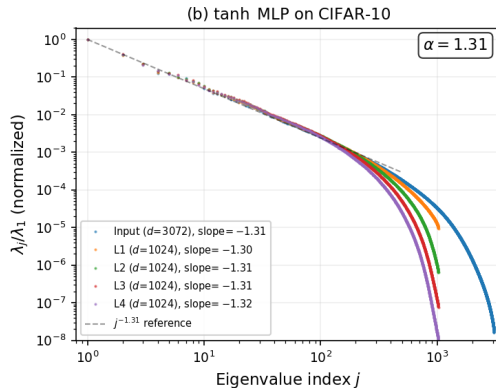
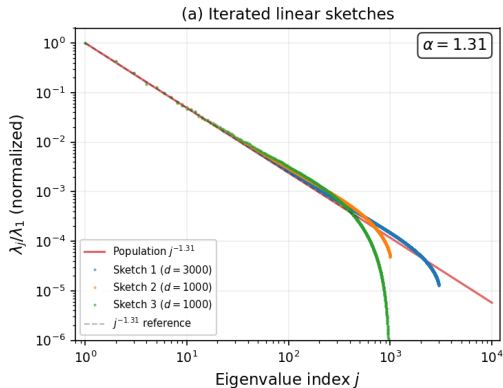
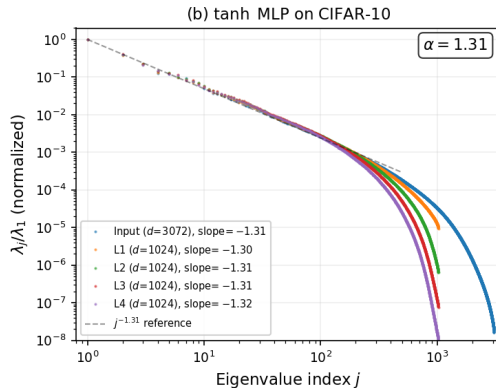
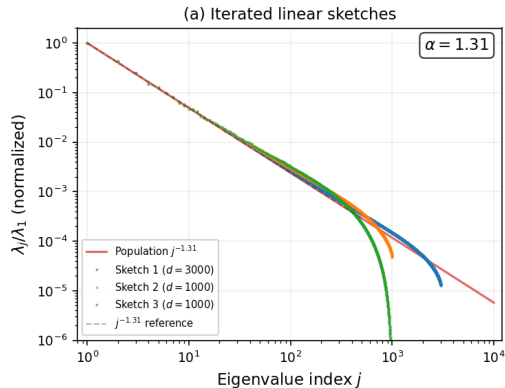


Image spectra survive random layers



The finite width changes the spectral cutoff much more than the fitted slope.

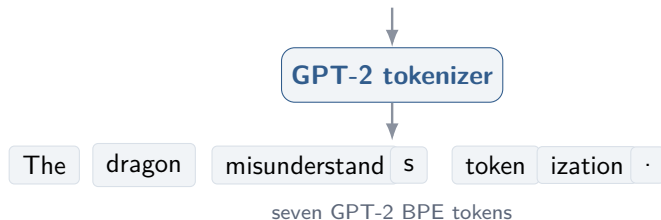
One sentence, three random vectors

The dragon misunderstands tokenization.

Start with a sentence.

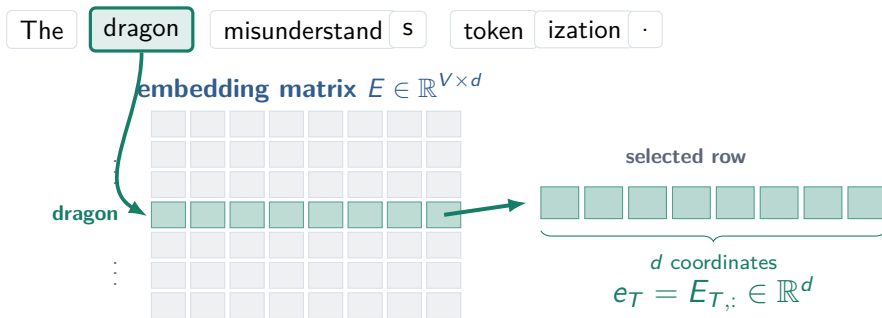
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Tokenization chooses the discrete symbols the model will see.

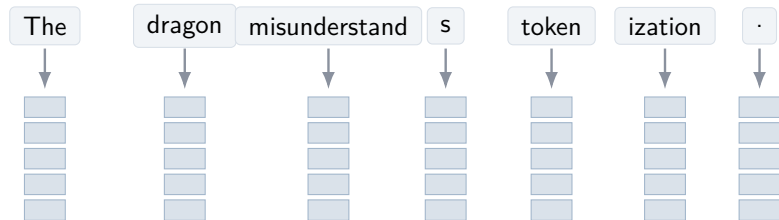
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For $T = \text{dragon}$, take one row of E : this is e_T .

One sentence, three random vectors

A context is a sequence of token embeddings

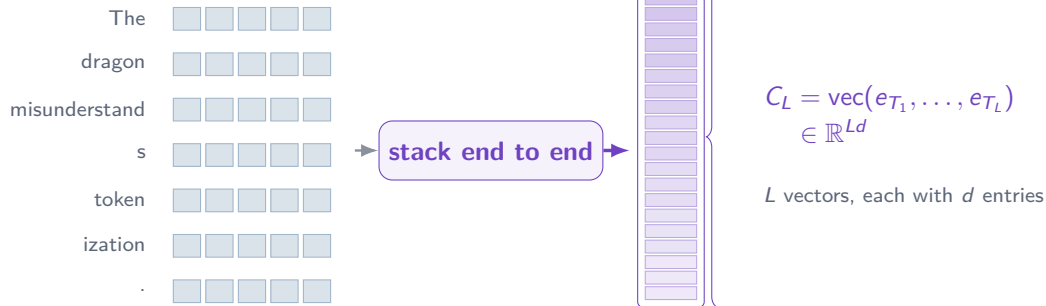


$e_{T_1}, e_{T_2}, \dots, e_{T_L} \in \mathbb{R}^d$
one d -dimensional vector for every token

Keeping the vectors separate remembers which embedding belongs to which token.

One sentence, three random vectors

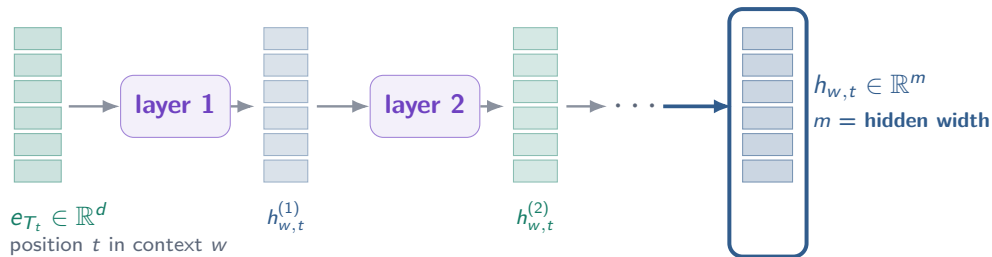
Vectorize the whole context



Stacking the window produces a second random vector, C_L .

One sentence, three random vectors

The network creates new random vectors



After each transformation, the activation $h_{w,t}$ is another random vector.

Zipf frequencies determine token covariance

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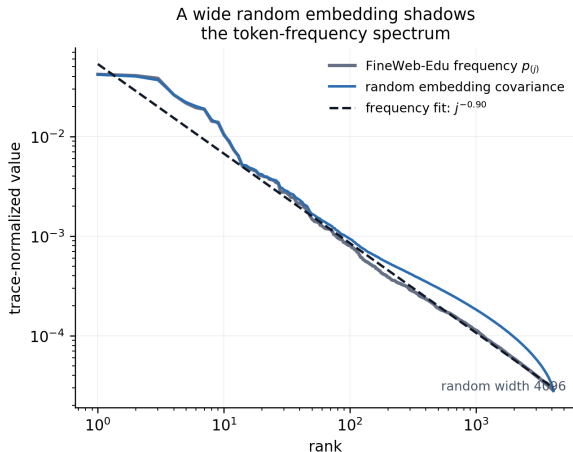
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A wide random embedding sketches this covariance.

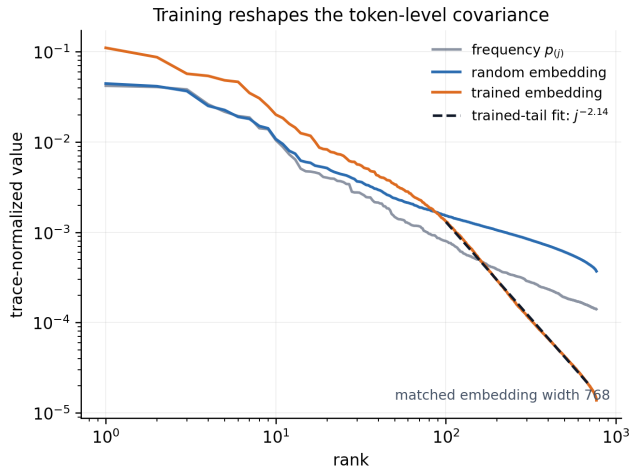
At initialization, frequency geometry survives the embedding



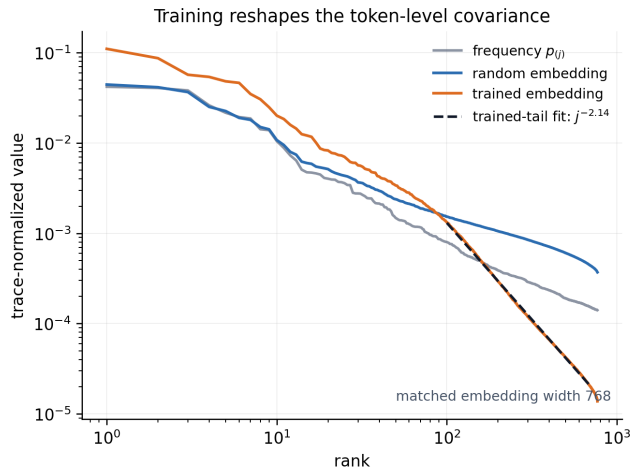
Held-out FineWeb-Edu tokens.

Training introduces a second spectral regime

- The trained head is reshaped by semantic overlap between token embeddings.

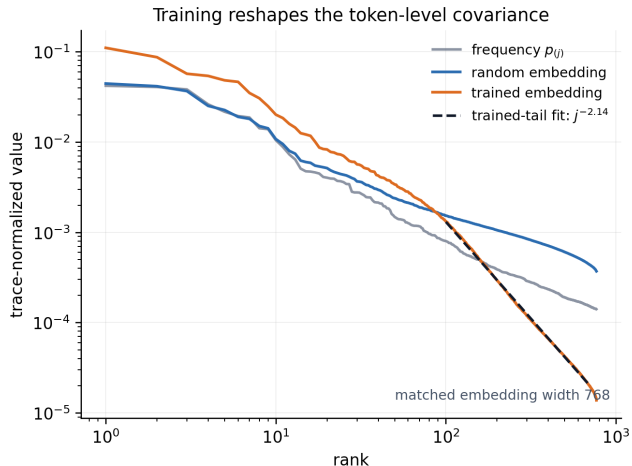


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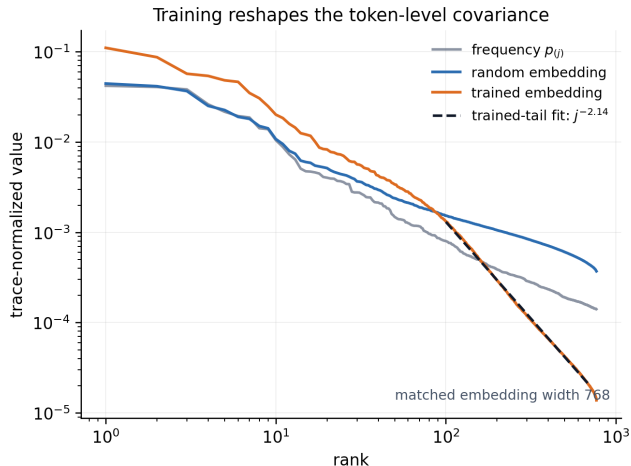
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Piecewise power laws give a tractable refinement of a single-exponent model.

Context covariance exposes token dependence

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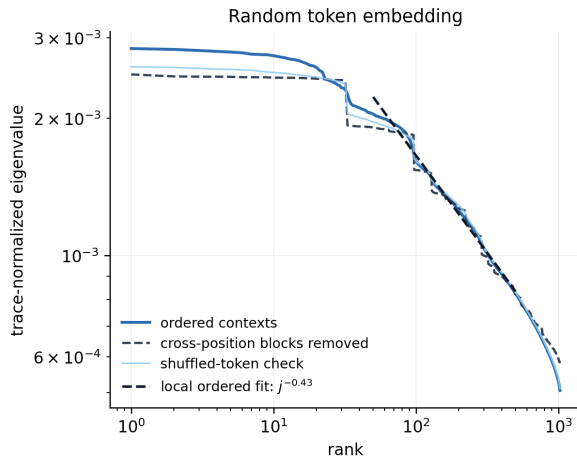
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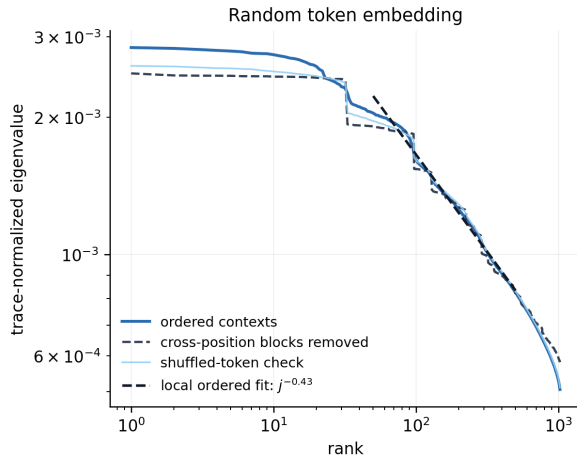
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- Γ_{aa} : token covariance at position a ;
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- independent tokens would make every off-diagonal block vanish.

Context spectra begin with a flatter phase

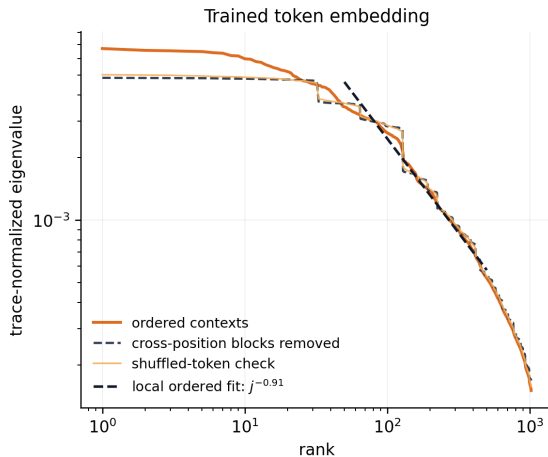


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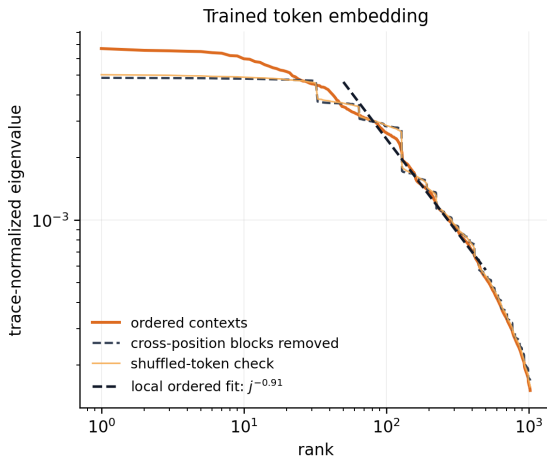


The initial flat regime gives way to a distinctly different spectral tail.

Training changes the context-spectrum crossover



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The trained representation again shows more than one spectral regime.

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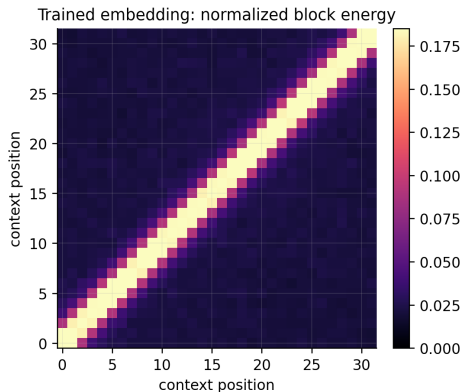
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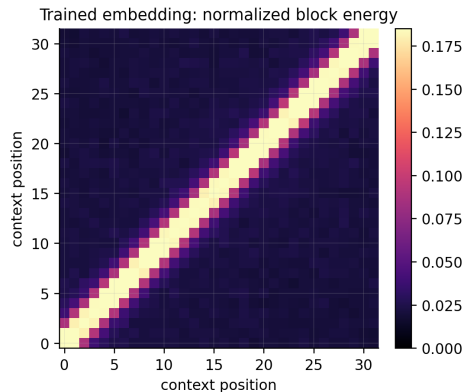
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Permuting token positions can completely rearrange the block geometry while leaving every eigenvalue unchanged.

The spectrum does not show this positional structure

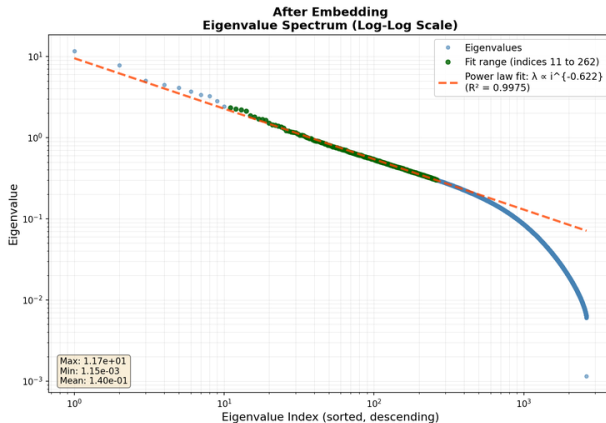


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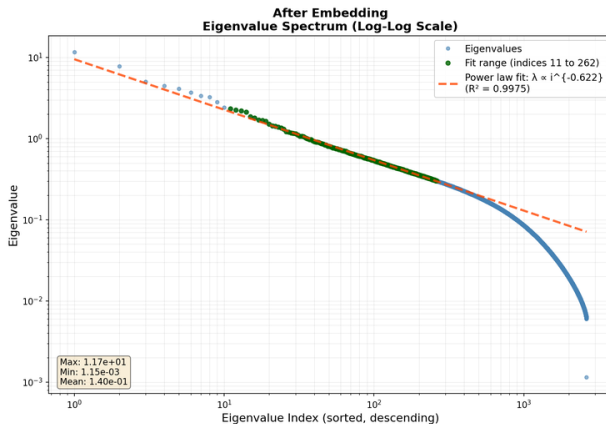


Permuting positions moves the diagonal band without changing the eigenvalues.

The embedding begins with an extended power-law regime

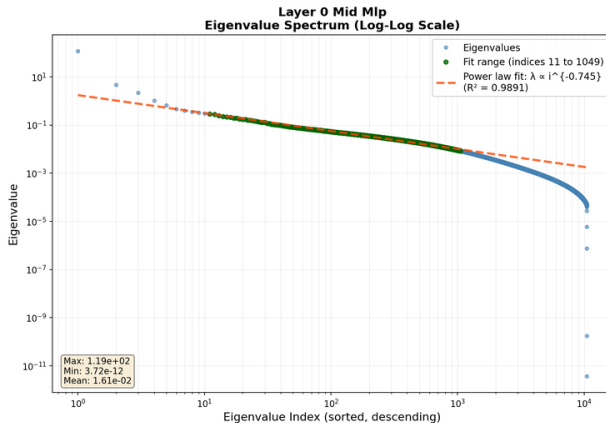


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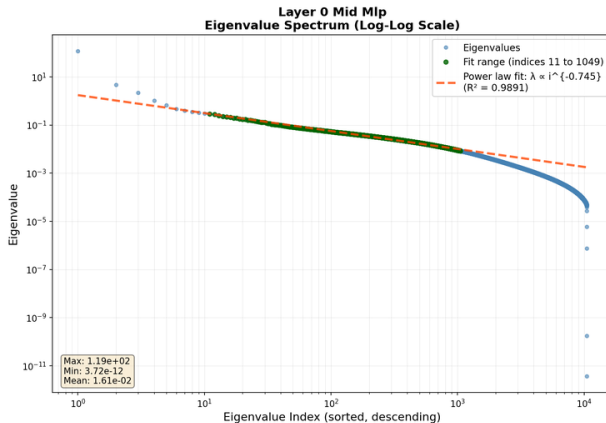


After the embedding, the fitted spectral exponent is approximately 0.62.

The power-law regime survives the first MLP

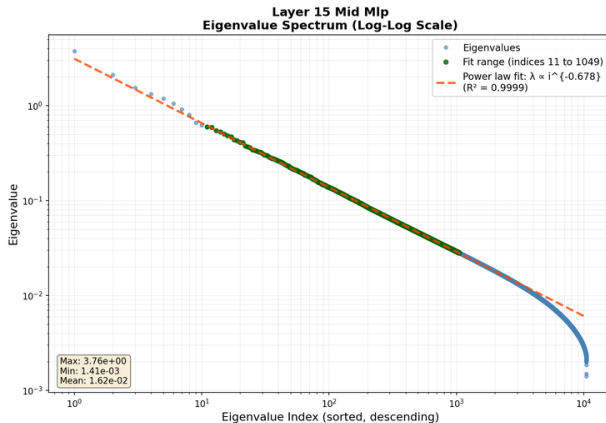


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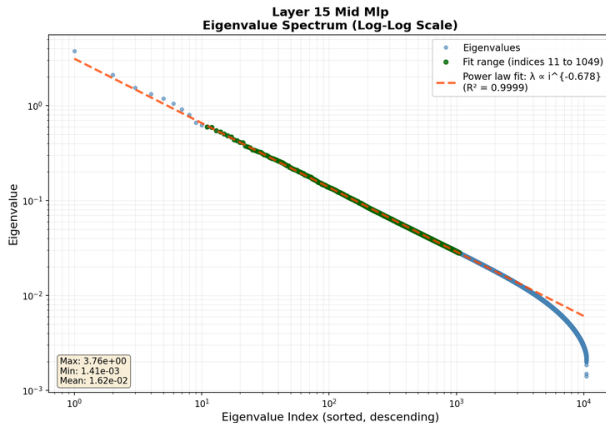


Inside layer 0, the fitted spectral exponent is approximately 0.75.

The same structure persists deep in the network

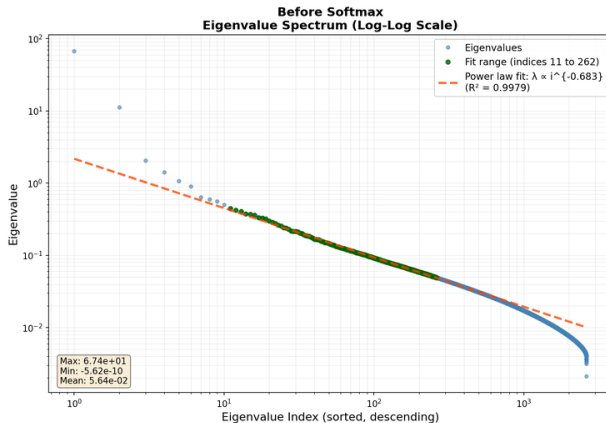


The same structure persists deep in the network

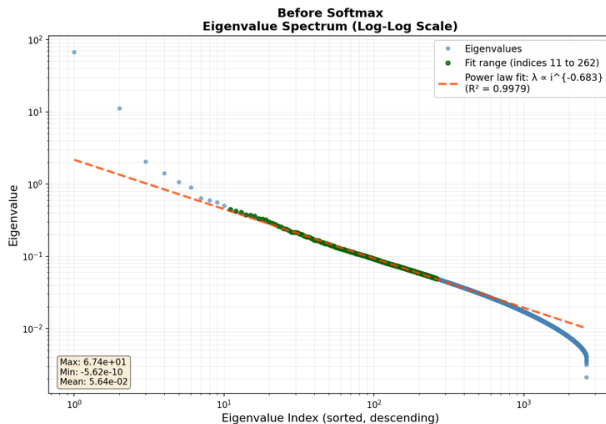


Inside layer 15, the fitted spectral exponent is approximately 0.68.

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Immediately before the softmax, the fitted spectral exponent is approximately 0.68.

Theory: weak preservation

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Constants, the eigenbasis, and logarithmic corrections may change while the polynomial exponent survives.

One-block stability can propagate through depth

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A uniform one-block statement propagates through finite depth by induction. The hard step is controlling one nonlinear transformation.

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Maloney, Roberts, and Sully used this setting to identify approximate spectral preservation in a solvable scaling model (Maloney et al. 2022).

The exponent survives monomial random features

Theorem: power-law preservation for monomial random features

Let $X \sim N(0, \Sigma)$ with $\lambda_j(\Sigma) \asymp j^{-\alpha}$. Let W have independent Gaussian entries and let $f(y) = y^p$. Then, with high probability over W ,

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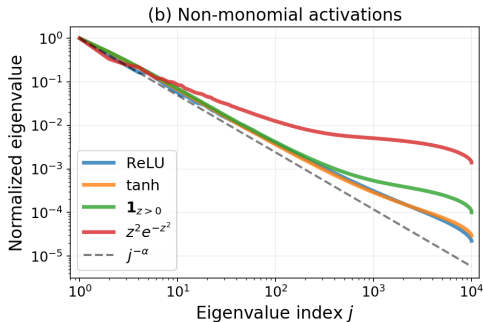
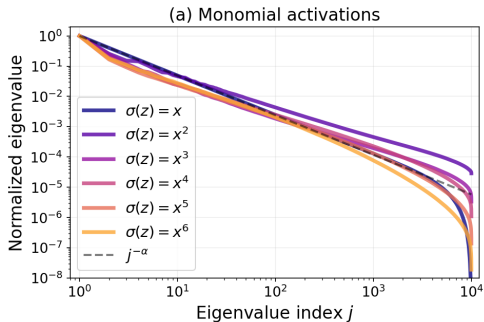
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P.-Ke Liang Xiao–Yizhe Zhu (2026): polynomial rate preserved; logarithmic correction added.

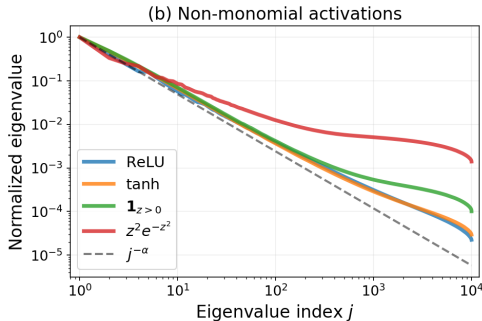
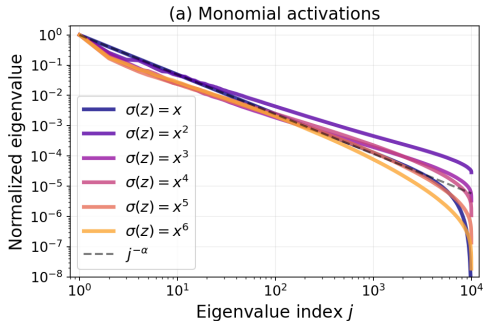
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Median postactivation eigenvalue, 100 trials ($\alpha = 1.31$, $v = d = 10000$, $m = 100000$)



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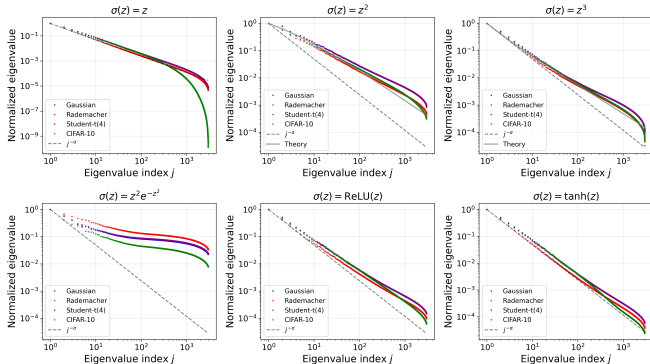
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The theorem covers monomials. ReLU, tanh, and other activations show related behavior numerically—but not every activation preserves the same tail.

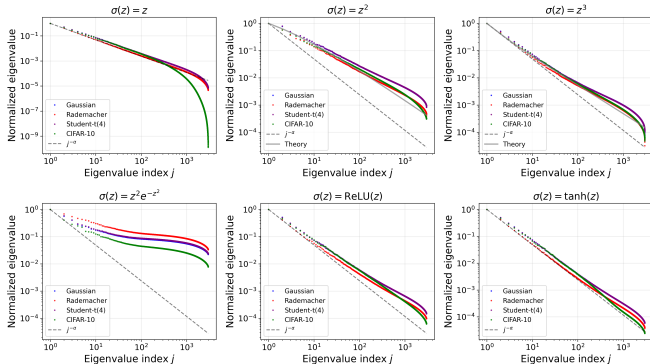
Preservation also appears beyond Gaussian inputs

Universality of Eigenvalue Spectra, $\alpha = 1.31$
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Similar slopes across four input distributions suggest a broader universality phenomenon.

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At $j = 3,000$

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The measured slope is 12.5% flatter.

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At realistic widths, the logarithm behaves like a different effective exponent.

The proof is a tensor covariance argument

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For the quadratic activation, every step is already visible in the second Wiener chaos.

Centering isolates the pure second chaos

Write $W = [w_1 \cdots w_d] \in \mathbb{R}^{\nu \times d}$, so $w_a \in \mathbb{R}^{\nu}$ for $a \in [d] := \{1, \dots, d\}$. Set

$$g_a = w_a^\top X, \quad q_a = \mathbb{E}_X[g_a^2] = w_a^\top \Sigma w_a.$$

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If $G = W^\top \Sigma W$ and $q = \text{diag}(G)$, then

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The two matrices differ by one rank-one term; interlacing preserves the tail.

Quadratic features live in symmetric tensor space

Let $X = \Sigma^{1/2}Z$, where $Z \sim N(0, I_V)$, and let $h(Z)$ collect the orthonormal basis functions $(Z_r^2 - 1)/\sqrt{2}$ and $Z_r Z_s$ for $r < s$. Then

$$\begin{aligned} :g_a^2: &:= \langle \phi_a, h(Z) \rangle, \\ \phi_a &= \left((\sqrt{2} \lambda_r w_{ra}^2)_r, (2\sqrt{\lambda_r \lambda_s} w_{ra} w_{sa})_{r < s} \right). \end{aligned}$$

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Thus, with $\Phi = [\phi_1 \ \cdots \ \phi_d] \in \mathbb{R}^{v(v+1)/2 \times d}$,

$$\boxed{K_{2,W}^{\text{chaos}} = \frac{1}{d} \Phi^\top \Phi}$$

Tensor eigenvalues are pair products

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Thus the spectral counting problem is a lattice-point problem under a hyperbola.

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Hence $j \asymp R \log R$, so

$$\lambda_j \asymp \left(\frac{\log(j+1)}{j} \right)^\alpha.$$

Width becomes sample size in tensor space

The nonzero eigenvalues of

$$\frac{1}{d}\Phi^\top\Phi \quad \text{and} \quad \hat{C}_{\text{chaos}} = \frac{1}{d}\Phi\Phi^\top = \frac{1}{d}\sum_{a=1}^d \phi_a\phi_a^\top$$

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On a k -dimensional spectral head, the desired estimate is

$$\left\| \frac{1}{d} \sum_{a=1}^d \xi_a \xi_a^\top - I_k \right\| \leq \delta.$$

The tensor features are heavy-tailed

Coordinates of ξ are products $w_r w_s$. For a unit vector u , Gaussian hypercontractivity gives

$$\|\langle \xi, u \rangle\|_{L^q} \leq (q-1) \|\langle \xi, u \rangle\|_{L^2}, \quad q \geq 2.$$

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For degree p , the current argument gives a sharp range

$$k_\star \asymp \frac{d}{\log^{p+1} d}; \quad p = 2 : \quad k_\star \asymp \frac{d}{\log^3 d}.$$

Two logarithms come from different mechanisms

Structural

$$\lambda_j \asymp \left(\frac{\log j}{j} \right)^\alpha$$

Comes from counting pair products

$$rs \leq R.$$

It is already present in the **population tensor spectrum**.

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Statistical

$$k_\star \asymp \frac{d}{\log^3 d}$$

Comes from estimating a covariance with
heavy-tailed tensor samples.

It limits the current **finite-width proof range**.

Open problems

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- **Sequences and attention.** If dependence decays along the sequence, what structure survives attention and depth?

Selected references

1. Elliot Paquette, Ke Liang Xiao, and Yizhe Zhu. *Power-Law Spectrum of the Random Feature Model* (2026). arXiv:2603.14578.

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doi:10.1007/s00440-025-01412-6.

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3. Alexander Maloney, Daniel A. Roberts, and James Sully. *A Solvable Model of Neural Scaling Laws* (2022). arXiv:2210.16859.

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In Module 2, we will explore how spectra affect learning curves.

We will introduce how this spectral structure can affect scaling laws.